```
Client.joins(:orders).where(orders: { created_at: time_range })
```

This will find all clients who have orders that were created yesterday, again using a `BETWEEN` SQL expression.

13 Eager Loading Associations

Eager loading is the mechanism for loading the associated records of the objects returned by `Model.find` using as few queries as possible.

N + 1 queries problem

Consider the following code, which finds 10 clients and prints their postcodes:

```
clients = Client.limit(10)

clients.each do |client|
  puts client.address.postcode
end
```

This code looks fine at the first sight. But the problem lies within the total number of queries executed. The above code executes 1 (to find 10 clients) + 10 (one per each client to load the address) = **11**queries in total.

Solution to N + 1 queries problem

Active Record lets you specify in advance all the associations that are going to be loaded. This is possible by specifying the `includes` method of the `Model.find` call. With `includes`, Active Record ensures that all of the specified associations are loaded using the minimum possible number of queries.

Revisiting the above case, we could rewrite `Client.limit(10)` to use eager load addresses:

```
clients = Client.includes(:address).limit(10)

clients.each do |client|
  puts client.address.postcode
end
```

The above code will execute just **2** queries, as opposed to **11** queries in the previous case:

```
SELECT * FROM clients LIMIT 10
SELECT addresses.* FROM addresses
  WHERE (addresses.client_id IN (1,2,3,4,5,6,7,8,9,10))
```

13.1 Eager Loading Multiple Associations

Active Record lets you eager load any number of associations with a single `Model.find` call by using an array, hash, or a nested hash of array/hash with the `includes` method.

13.1.1 Array of Multiple Associations

```
Article.includes(:category, :comments)
```

This loads all the articles and the associated category and comments for each article.

13.1.2 Nested Associations Hash

```
Category.includes(articles: [{ comments: :guest }, :tags]).find(1)
```

This will find the category with id 1 and eager load all of the associated articles, the associated articles' tags and comments, and every comment's guest association.

13.2 Specifying Conditions on Eager Loaded Associations

Even though Active Record lets you specify conditions on the eager loaded associations just like `joins`, the recommended way is to use joins instead.

However if you must do this, you may use `where` as you would normally.